

Wave Function

Quantum Mechanics

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1 Introduction

These notes summarize the foundational concepts of quantum mechanics as presented in the first chapter of a standard textbook. The core topics include the Schrödinger equation, the statistical interpretation of the wave function, probability theory, normalization, momentum, and the uncertainty principle.

2 The Schrödinger Equation

2.1 Classical vs. Quantum Description

- **Classical mechanics:** For a particle of mass m constrained to move along the x -axis under a force $F(x, t)$, the goal is to determine the position $x(t)$. Using Newton's second law $F = ma$, and for conservative systems where $F = -\partial V/\partial x$ (with V being the potential energy), we solve:

$$m \frac{d^2 x}{dt^2} = -\frac{\partial V}{\partial x}.$$

Initial conditions (position and velocity at $t = 0$) determine $x(t)$.

- **Quantum mechanics:** The state of the particle is described by a **wave function** $\Psi(x, t)$. The evolution of $\Psi(x, t)$ is governed by the **Schrödinger equation**.

2.2 The Time-Dependent Schrödinger Equation

For a particle of mass m in one dimension with potential energy $V(x, t)$, the Schrödinger equation is:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi. \quad (1)$$

Here:

- i is the imaginary unit ($i^2 = -1$).
- $\hbar$ is the reduced Planck's constant:

$$\hbar = \frac{h}{2\pi} = 1.054573 \times 10^{-34} \text{ J}\cdot\text{s}.$$

Given the initial wave function $\Psi(x, 0)$, the Schrödinger equation determines $\Psi(x, t)$ for all future times, analogous to Newton's second law in classical mechanics.

3 The Statistical Interpretation

3.1 Born's Interpretation

The wave function itself is spread out in space, but a particle is localized. The connection is provided by **Born's statistical interpretation**:

$$|\Psi(x, t)|^2 dx = \text{Probability of finding the particle between } x \text{ and } (x + dx) \text{ at time } t. \quad (2)$$

Thus $|\Psi(x, t)|^2$ is a **probability density**.

3.2 Indeterminacy and Interpretations

The statistical interpretation introduces fundamental indeterminacy: even knowing Ψ completely, one cannot predict the outcome of a position measurement with certainty, only its probability. Three main philosophical positions have been historically held:

1. **Realist (Einstein):** The particle *was* at a specific location before measurement. Quantum mechanics is incomplete; **hidden variables** might provide a complete description.
2. **Orthodox (Copenhagen interpretation, Bohr):** The particle was not anywhere until measured. The act of measurement forces the particle to "take a stand."
3. **Agnostic (Pauli):** Refuse to answer, as the question is metaphysical and untestable.

Bell's theorem (1964) showed that the realist position makes experimentally testable predictions different from the orthodox view. Experiments have decisively confirmed the orthodox interpretation: a particle does not have a precise position prior to measurement.

3.3 Wave Function Collapse

Upon measuring the position and finding the particle at point C , the wave function **collapses** discontinuously to a spike at C . Immediate repeated measurements will yield the same value C . Subsequently, the wave function evolves again according to the Schrödinger equation.

Thus, there are two distinct processes:

- **Ordinary evolution:** Continuous, deterministic evolution via the Schrödinger equation.
- **Measurement:** Sudden, discontinuous collapse of the wave function.

4 Probability Theory Fundamentals

4.1 Discrete Distributions

For a discrete variable j with probability $P(j) = N(j)/N$:

- Normalization:

$$\sum_j P(j) = 1.$$

- Expectation value (average):

$$\langle j \rangle = \sum_j j P(j).$$

- Expectation of a function $f(j)$:

$$\langle f(j) \rangle = \sum_j f(j) P(j).$$

- Variance and standard deviation:

$$\sigma^2 \equiv \langle (\Delta j)^2 \rangle = \langle (j - \langle j \rangle)^2 \rangle = \langle j^2 \rangle - \langle j \rangle^2.$$

σ is the standard deviation, measuring the spread around the mean.

4.2 Continuous Distributions

For a continuous variable x with probability density $\rho(x)$:

- Probability of finding x in $[a, b]$:

$$P_{ab} = \int_a^b \rho(x) dx.$$

- Normalization:

$$\int_{-\infty}^{+\infty} \rho(x) dx = 1.$$

- Expectation values:

$$\langle x \rangle = \int_{-\infty}^{+\infty} x \rho(x) dx, \quad \langle f(x) \rangle = \int_{-\infty}^{+\infty} f(x) \rho(x) dx.$$

- Variance:

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2.$$

5 Normalization of the Wave Function

5.1 Condition for Physical States

Since $|\Psi(x, t)|^2$ is a probability density, it must satisfy:

$$\int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = 1 \quad \text{for all } t. \quad (3)$$

This is the **normalization condition**. Any solution to the Schrödinger equation can be multiplied by a constant; we choose this constant to satisfy Eq. (3). Solutions for which the integral diverges or which are identically zero are non-normalizable and do not represent physical particles.

5.2 Preservation of Normalization

It is crucial that if Ψ is normalized at $t = 0$, it remains normalized for all time. Proof:

$$\begin{aligned} \frac{d}{dt} \int_{-\infty}^{+\infty} |\Psi|^2 dx &= \int_{-\infty}^{+\infty} \frac{\partial}{\partial t} |\Psi|^2 dx, \\ \frac{\partial}{\partial t} |\Psi|^2 &= \Psi^* \frac{\partial \Psi}{\partial t} + \frac{\partial \Psi^*}{\partial t} \Psi. \end{aligned}$$

Using the Schrödinger equation and its complex conjugate:

$$\begin{aligned} \frac{\partial \Psi}{\partial t} &= \frac{i\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} - \frac{i}{\hbar} V \Psi, \\ \frac{\partial \Psi^*}{\partial t} &= -\frac{i\hbar}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} + \frac{i}{\hbar} V \Psi^*, \end{aligned}$$

we obtain:

$$\frac{\partial}{\partial t} |\Psi|^2 = \frac{\partial}{\partial x} \left[\frac{i\hbar}{2m} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \frac{\partial \Psi^*}{\partial x} \Psi \right) \right].$$

Integrating over all x :

$$\frac{d}{dt} \int_{-\infty}^{+\infty} |\Psi|^2 dx = \frac{i\hbar}{2m} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \frac{\partial \Psi^*}{\partial x} \Psi \right) \Bigg|_{-\infty}^{+\infty}.$$

For a normalizable wave function, $\Psi \rightarrow 0$ as $x \rightarrow \pm\infty$, so the boundary term vanishes. Hence:

$$\frac{d}{dt} \int_{-\infty}^{+\infty} |\Psi|^2 dx = 0,$$

and the normalization is conserved.

6 Momentum and Expectation Values

6.1 Expectation of Position

The expectation value of position for a particle in state Ψ is:

$$\langle x \rangle = \int_{-\infty}^{+\infty} x |\Psi(x, t)|^2 dx = \int \Psi^* x \Psi dx. \quad (4)$$

This is the average of position measurements on an ensemble of identically prepared systems.

6.2 Expectation of Momentum

The time derivative of $\langle x \rangle$ gives:

$$\frac{d\langle x \rangle}{dt} = \int x \frac{\partial}{\partial t} |\Psi|^2 dx = \frac{i\hbar}{2m} \int x \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \frac{\partial \Psi^*}{\partial x} \Psi \right) dx.$$

Integration by parts (assuming $\Psi \rightarrow 0$ at infinities) yields:

$$\frac{d\langle x \rangle}{dt} = -\frac{i\hbar}{2m} \int \left(\Psi^* \frac{\partial \Psi}{\partial x} - \frac{\partial \Psi^*}{\partial x} \Psi \right) dx.$$

A second integration by parts on the second term gives:

$$\frac{d\langle x \rangle}{dt} = -\frac{i\hbar}{m} \int \Psi^* \frac{\partial \Psi}{\partial x} dx.$$

Defining the expectation value of momentum as $\langle p \rangle = m \frac{d\langle x \rangle}{dt}$, we obtain:

$$\langle p \rangle = -i\hbar \int \Psi^* \frac{\partial \Psi}{\partial x} dx = \int \Psi^* \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \Psi dx. \quad (5)$$

6.3 Operators

In quantum mechanics:

- The **position operator** is $\hat{x} = x$.
- The **momentum operator** is $\hat{p} = \frac{\hbar}{i} \frac{\partial}{\partial x}$.

For any dynamical variable $Q(x, p)$, the expectation value is:

$$\langle Q(x, p) \rangle = \int \Psi^* Q \left(x, \frac{\hbar}{i} \frac{\partial}{\partial x} \right) \Psi dx. \quad (6)$$

For example, kinetic energy $T = p^2/(2m)$:

$$\langle T \rangle = \frac{-\hbar^2}{2m} \int \Psi^* \frac{\partial^2 \Psi}{\partial x^2} dx.$$

6.4 Ehrenfest's Theorem

The expectation values obey Newton's second law:

$$\frac{d\langle p \rangle}{dt} = \left\langle -\frac{\partial V}{\partial x} \right\rangle. \quad (7)$$

This is known as **Ehrenfest's theorem**.

7 The Uncertainty Principle

7.1 Wave-Particle Duality and the de Broglie Relation

For a quantum particle, the de Broglie relation connects momentum p to wavelength λ :

$$p = \frac{h}{\lambda} = \frac{2\pi\hbar}{\lambda}. \quad (8)$$

A wave with a well-defined wavelength (momentum) is spread out in space; a localized wave packet contains a range of wavelengths.

7.2 Heisenberg's Uncertainty Principle

There is a fundamental trade-off between the precision of position and momentum measurements. Quantitatively:

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}, \quad (9)$$

where σ_x and σ_p are the standard deviations of position and momentum, respectively.

Interpretation: If many identically prepared particles are measured for position, the results cluster around $\langle x \rangle$ with spread σ_x . On the same ensemble, momentum measurements yield spread σ_p . The product $\sigma_x \sigma_p$ cannot be smaller than $\hbar/2$. Preparing a state with a very small σ_x (highly localized) forces σ_p to be large, and vice versa. This is not due to measurement disturbance but is an inherent property of quantum states.

8 Key Equations Summary

$$\text{Schrödinger equation: } i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi. \quad (10)$$

$$\text{Born interpretation: } |\Psi|^2 dx = \text{Probability in } [x, x + dx]. \quad (11)$$

$$\text{Normalization: } \int_{-\infty}^{+\infty} |\Psi|^2 dx = 1. \quad (12)$$

$$\text{Position expectation: } \langle x \rangle = \int \Psi^* x \Psi dx. \quad (13)$$

$$\text{Momentum expectation: } \langle p \rangle = \int \Psi^* \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \Psi dx. \quad (14)$$

$$\text{Uncertainty principle: } \sigma_x \sigma_p \geq \frac{\hbar}{2}. \quad (15)$$